

# Fairness-Aware Service Allocation in Adaptive Quantum Networks: Risks, Metrics, and Auditable Mediation

Piter Garcia

Focus Area: Fairness

Capstone: Fairness-Aware Adaptive Routing for Quantum Networks

## I. SYSTEM CONTEXT AND FAIRNESS SCOPE

The artifact under analysis is a threat-aware quantum-routing controller with fairness-evaluation and mediation components. At each decision round, it observes a possibly incomplete network state, uses a bandit policy to select an entanglement route, and allocates limited quantum capacity under stochastic links, partial feedback, and configurable threats. Link quality, memory constraints, and attacks affect route quality and realized performance; allocator choice shapes qubit distribution, while replay capacity controls how much experience replay-based policies retain for learning. These decisions are consequential because entanglement distribution depends on probabilistic link success, scarce resources, and competition among requests [1], [2]. Existing components support post-run group evaluation, and a quantum-specific mediation contract exists for future live integration.

Fairness is allocative: strong network-wide efficiency can coexist with systematically worse service for some flows. The controller affects whether requests succeed, their latency and retry burden, and whether repeated allocations cause starvation. Stakeholders include competing applications or tenants, operators responsible for allocation policy, and downstream users of the resulting entanglement paths. Prespecified equity units may include latency-sensitive versus delay-tolerant traffic, high- versus low-demand requests, or high- versus low-context-quality cohorts. These are proposed operational groups, not demographic or validated cohorts, and must not be defined after observing outcomes. If service classes become associated with people or institutions, protected-attribute analysis and stakeholder review would also be necessary. Quantum-routing research treats finite capacity and simultaneous requests as fair-allocation problems, supporting analysis beyond aggregate throughput [3], [4].

## II. FAIRNESS RISK ANALYSIS

Bias can enter throughout the routing pipeline. Following established life-cycle taxonomies [5], [6], representation harm arises when workloads omit difficult topologies, low-context flows, or relevant service classes. Measurement harm arises when the quantum trace lacks demand, allocation, route, fidelity, retry, starvation, or context-quality evidence needed to interpret group-conditioned outcomes. Aggregation harm occurs when one reward function treats different latency, fidelity, or demand

requirements as interchangeable. Evaluation harm occurs when aggregate throughput or reward conceals class-level outcomes, and deployment harm arises when a policy is transferred across materially different topologies, demand distributions, or threat regimes.

Sequential learning can compound these disparities because a bandit observes rewards only for selected actions. Unequal selection produces unequal evidence; depending on exploration, reward estimates, context quality, and threats, that imbalance can influence later decisions. Fair-bandit research shows that fairness constraints alter exploration and regret rather than acting as cost-free post-processing [7]; contextual-allocation work likewise motivates explicit minimum-service concerns [8]. Incomplete or stale context can therefore disadvantage a flow without any protected attribute, while nonstationarity may concentrate failures, retries, or delay on one class despite acceptable aggregate performance.

Plausible harms include longer waits for latency-sensitive traffic, demand-normalized under-service of high-demand flows, and poorer decisions for weak-context flows. Long or low-fidelity routes may be repeatedly deprioritized until requests starve. These are allocative harms because access to a scarce capability is degraded, reliability harms when failures or delay persist, and accountability harms when operators cannot reconstruct why a flow received inferior service.

## III. FAIRNESS OBJECTIVES, MITIGATIONS, AND VALIDATION

Classical criteria are useful reference points but do not transfer mechanically to quantum-resource allocation. Demographic parity supplies an analogous test of access or success rates across groups; equalized odds and equality of opportunity concern prediction errors conditional on outcomes [9]. Calibration—agreement between predicted and observed success—and parity may conflict under different base rates, and desirable criteria cannot generally be satisfied together [10], [11]. Equal raw allocation may also conflict with equal success when demand and path quality differ. The primary objective is therefore demand-normalized service equity, supplemented by group outcomes and a starvation constraint.

For group  $g$  over horizon  $T$ , define

$$A_g(T) = \frac{\sum_{t=1}^T Q_{g,t}}{\sum_{t=1}^T D_{g,t}}, \quad \Delta_A(T) = \max_{g,h} |A_g(T) - A_h(T)|, \quad (1)$$

where  $Q_{g,t}$  is allocated capacity and  $D_{g,t}$  is declared demand. Audits should also report success and latency gaps, Jain's index over normalized access, worst-group outcomes, retries, starvation, and supported fidelity failures [12]. Experiments must prespecify group definitions, denominators, zero-demand exclusions, missing-context rules, and physical-infeasibility conditions.

The first mitigation is an enriched decision trace containing group, demand, context quality, candidate and selected routes, policy scores, allocated capacity, fidelity, success, latency, retries, and starvation age. A bounded, auditable intervention at the policy-score or allocator boundary could use

$$S_t^{\text{fair}}(a) = U_t(a) - \lambda[\hat{\Delta}_t(a) - \epsilon]_+ - \gamma C_t(a), \quad (2)$$

where  $U_t(a)$  is utility,  $\hat{\Delta}_t(a)$  estimated disparity,  $\epsilon$  tolerated slack, and  $C_t(a)$  a missing-context penalty. Under this proposed formulation and without a hard constraint,  $\lambda = \gamma = 0$  recovers the base score and defines the unmediated comparison condition; a feasible service floor or starvation limit may supplement the penalty [8].

Validation should compare mediated and unmediated configurations across matched policies, allocators, replay-capacity settings, topologies, threats, and random seeds. Reports should place utility, regret, success, and fidelity beside equity measures, worst-group outcomes, and uncertainty intervals. The prespecified utility–equity frontier, rather than a favorable post hoc operating point, should guide selection. Fidelity and safety constraints remain binding even when disparity penalties or service floors reduce throughput, increase other flows' latency, or spend resources on physically disadvantaged routes.

#### IV. BROADER IMPACTS

The broader aim is to prevent aggregate performance from becoming the sole objective when scarce quantum-network resources are allocated across competing flows. A service-equity layer can expose disparities and make intervention costs auditable. The claim is deliberately narrow: the project should measure flow-conditioned outcomes and evaluate a mediator and its utility–equity frontier; it does not establish that a network or allocation policy is socially fair.

#### LIMITATIONS

Live fairness-mediated quantum evaluation remains prospective. Fairness conclusions from simulation will be limited by modeled workloads, topologies, physical assumptions, and threats. Service classes are engineering choices, demand normalization embeds assumptions, criteria can conflict, and small classes can yield unstable estimates. Repeated trials and uncertainty intervals are therefore necessary; insufficient evidence must be reported rather than treated as absence of disparity, and any service floor must be justified rather

than assumed fair. Mediation may optimize an imperfect proxy or reduce utility. Deployment would require stakeholder review, hardware-grounded validation, transparent class and denominator rules, authority over alerts and weights, a rollback procedure, and continuing audits. Manipulation of threat or routing signals and exposure of flow telemetry require separate security and privacy analysis.

#### REFERENCES

- [1] S. Wehner, D. Elkouss, and R. Hanson, "Quantum internet: A vision for the road ahead," *Science*, vol. 362, no. 6412, p. eaam9288, 2018.
- [2] M. Pant, H. Krovi, D. Towsley, L. Tassioulas, L. Jiang, P. Basu, D. Englund, and S. Guha, "Routing entanglement in the quantum internet," *npj Quantum Information*, vol. 5, p. 25, 2019.
- [3] C. Li, T. Li, Y.-X. Liu, and P. Cappellaro, "Effective routing design for remote entanglement generation on quantum networks," *npj Quantum Information*, vol. 7, no. 1, p. 10, 2021.
- [4] C. Cicconetti, M. Conti, and A. Passarella, "Service differentiation and fair sharing in distributed quantum computing," *Pervasive and Mobile Computing*, vol. 90, p. 101758, 2023.
- [5] H. Suresh and J. Gutttag, "A framework for understanding sources of harm throughout the machine learning life cycle," in *Proceedings of the 1st ACM Conference on Equity and Access in Algorithms, Mechanisms, and Optimization*, 2021.
- [6] N. Mehrabi, F. Morstatter, N. Saxena, K. Lerman, and A. Galstyan, "A survey on bias and fairness in machine learning," *ACM Computing Surveys*, vol. 54, no. 6, pp. 1–35, 2021.
- [7] M. Joseph, M. Kearns, J. Morgenstern, and A. Roth, "Fairness in learning: Classic and contextual bandits," in *Advances in Neural Information Processing Systems*, vol. 29, 2016.
- [8] Y. Chen, A. Cuellar, H. Luo, J. Modi, H. Nemlekar, and S. Nikolaidis, "Fair contextual multi-armed bandits: Theory and experiments," in *Proceedings of the 36th Conference on Uncertainty in Artificial Intelligence*, ser. Proceedings of Machine Learning Research, vol. 124, 2020, pp. 181–190.
- [9] M. Hardt, E. Price, and N. Srebro, "Equality of opportunity in supervised learning," in *Advances in Neural Information Processing Systems*, vol. 29, 2016.
- [10] A. Chouldechova, "Fair prediction with disparate impact: A study of bias in recidivism prediction instruments," *Big Data*, vol. 5, no. 2, pp. 153–163, 2017.
- [11] J. Kleinberg, S. Mullainathan, and M. Raghavan, "Inherent trade-offs in the fair determination of risk scores," in *8th Innovations in Theoretical Computer Science Conference (ITCS 2017)*, ser. Leibniz International Proceedings in Informatics, vol. 67, 2017, pp. 43:1–43:23.
- [12] R. K. Jain, D.-M. W. Chiu, and W. R. Hawe, "A quantitative measure of fairness and discrimination for resource allocation in shared computer systems," Digital Equipment Corporation, Tech. Rep. DEC-TR-301, 1984.